J–C–94b
August 8, 1962

SUPERSEDING
Int. Fed. Spec. J-C-0094a (GSA-FSS)
September 1, 1961 and
Fed. Spec. J-C-94
July 13, 1955

FEDERAL SPECIFICATION

CABLE AND WIRE, POWER, ELECTRICAL, NONMETALLIC-SHEATHED

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers multiple-conductor, nonmetallic-sheathed cables and single- and multiple-conductor, underground feeder and branch circuit cables and wires, employing copper or aluminum, solid or stranded conductors, for use in wiring systems operating at a potential of 600 volts or less.

1.2 Classification.

1.2.1 *Types, classes, grades, and number of conductors.* Cable and wire furnished under this specification shall be of the following types, classes, grades, and number of conductors, as specified (see 6.1).

Types:

Type NM.—For use in dry or moist locations.

Type NMC.—For use in dry or moist and mildly corrosive locations.

Type UF.—For direct burial in earth.

Classes:

Class 1.—With uninsulated grounding conductor (Types NM and NMC only).

Class 2.—Without ground conductor.

Grades:

Grade I.—With RHW grade rubber insulation.

Grade II.—With TW grade thermoplastic insulation.

Grade III.—With THW grade thermoplastic insulation.

Number of insulated conductors:

1 conductor (Type UF only).
2 conductors.
3 conductors.
4 conductors.

1.2.2 *Conductor sizes.* Multiple-conductor cables, AWG 14-4 if of copper; AWG 12-4 if of aluminum. Single-conductor type UF wire AWG/14-4/0 if of copper; AWG 12-4/0 if of aluminum.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification.

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Federal Specifications:

J-C-98 —Cable and wire, Insulated; Methods of Sampling and Testing.
UU-P-271—Paper, Wrapping, Waterproofed, Kraft.
PPP-B-601—Boxes, Wood, Cleated-Plywood.
PPP-B-621—Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636—Box, Fiberboard.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D.C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specifications:

MIL-L-10547—Liner, Case, Waterproof.
MIL-C-12000—Cable, Cord and Wire, Electric Packaging of.

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109—Inspection Terms and Definitions.
MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following publication forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply:

Underwriters' Laboratories, Inc. Standard:

UL 719—Standard for Nonmetallic Sheathed and Underground Feeder Cables.

(Copies may be obtained upon application to Underwriters' Laboratories Inc. 161 Sixth Avenue, New York, N.Y.; 207 East Ohio Street, Chicago, Ill.; 1655 Scott Blvd., Santa Clara, Cal.)

3. REQUIREMENTS

3.1 Cable and wire covered by this specification shall conform to the requirements of Underwriters' Laboratories, Inc. Standard UL 719 for Nonmetallic-Sheathed and Underground Feeder Cables.

3.2 Fire and casualty hazards.

3.2.1 Each bidder shall submit to the contracting agency proof that the cable and wire he proposes to supply under this specification conforms to the requirements of Underwriters' Laboratories, Inc. Standard UL 719. The label or listing of the Underwriters' Laboratories, Inc. may be accepted as evidence that the cable conforms to these requirements.

3.2.2 In lieu of the Underwriters' Laboratories, Inc. label or listing, the bidder may submit independent proof satisfactory to the contracting agency that his cable and

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wire conforms to the published standards as determined by inspection and tests specified in the above referenced Underwriters' Laboratories, Inc. Standard UL 719.

3.2.3 Compliance with the above preliminary requirements in regard to fire and casualty hazards does not absolve the bidder from complete compliance with the requirements of this specification in order to secure acceptance of his material.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Supplier responsibilities for inspection. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure suppliers and services conform to prescribed requirements.

4.1.1 Unless otherwise specified, such inspections by the supplier may be performed on a statistical or production lot basis to assure that the quality of manufacture is equal to the standards set by Underwriters' Laboratories, Inc. (see 6.2).

4.2 Sampling, nonmilitary.

4.2.1 *Lot.* Unless otherwise specified in the invitation for bids, the lot shall be formed as described in paragraph 3 of section 6 of J-C-98.

4.2.2 *Sampling for inspection and tests.* Unless otherwise specified in the invitation for bids, the sample shall be taken as described in paragraph 6 of section 6 of J-C-98.

4.3 Visual inspection, nonmilitary.

4.3.1 Each inspection unit in the sample, 4.2.2, as well as the package from which the inspection unit is taken, shall be examined visually before any tests are made. Visual inspection shall cover construction, splices, marking, type of conductor, joints, and repairs, and may be limited to that part of the package which is visible without unrolling.

4.4 General inspection, Military.

4.4.1 *Definitions.* Definitions of inspection terms used herein shall conform to MIL-STD-109.

4.4.2 *Acceptance inspection.* Acceptance inspection shall consist of group A, B, and C inspection in 4.4.2.1 through 4.4.2.3.1. Group B inspection shall normally be performed on lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection. However, the order may be varied when the Government considers it more practical to select separate samples for group B or C inspection or both.

4.4.2.1 *Group A inspection.* Group A inspection including sampling shall be in accordance with table I and with MIL-STD-105. Group A inspection shall be performed in any order which is satisfactory to the Government inspector.

TABLE I. *Group A inspection*

Inspection	AQL	
	Major	Minor
Visual and mechanical:	*Percent*	*Percent*
General construction Splices Marking Type of conductor (diameter) Joints Repairs	1.0	4.0
Electrical:		
Dielectric strength	1.0[2]	[1]

[1]All electrical defects are considered major

[2]1.0 percent unless otherwise specified.

4.4.2.2 *Group B inspection.* This inspection including sampling shall conform to table II and to MIL-STD-105. Unless other-

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wise specified herein, normal inspection shall be used at the start of the contract. The reduced inspection procedure shall be R-1.

TABLE II. *Group B inspection*

Inspection	Sampling plan
Subgroup 1:	
Average thickness of insulation Minimum thickness of insulation (less than nominal 3/64 inch) Minimum thickness of insulation (nominal 3/64 inch and over) Thickness of Jacket, average and minimum Tangent of angle between braid and axis of cable Picks per inch Yarn size Carriers and ends per carrier, braid Length of lay, conductor	See 4.4.2.2.1
Subgroup 2:	
Accelerated aging Air oven test Oxygen pressure test Change in tensile strength after oil immersion	6 specimens per 100,000 feet
Subgroup 3:	
Flame resistance, insulation and jacket Moisture absorption, braid Dripping and flaking, braid	6 specimens per 100,000 feet or fraction thereof

4.4.2.2.1 *Group B sampling plans.* The group B sampling plans as listed in table II shall be as follows:

AQL	Inspection level for normal inspection	Inspection level for reduced inspection
Percent		
6.5	L-7	L-5

4.4.2.2.2 *Order of inspection within group B.* Group B inspection shall be performed in any order which is satisfactory to the Government inspector.

4.4.2.2.3 *Procedure in the case of failure.* If an inspection lot is rejected, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken. If the contractor and Government inspector cannot agree on the effectiveness of the corrective action, the matter shall be referred to the contracting officer for resolution.

4.4.2.3 *Group C inspection.* This inspection shall be as listed in table III.

TABLE III. *Group C inspection*

Inspection	Sampling plan
Flexibility Impact resistance Crush resistance Tensile strength Elongation Overload and high temperature flow Mechanical water absorption Specific inductive capacity	6 specimens per 500,000 feet

4.4.2.3.1 *Specimens.* Six specimens shall be prepared and tested from each lot of 500,000 feet, or fraction thereof, of cable and wire. If each of the 6 specimens meet the specified requirements, accept the lot.

4.4.2.3.1.1 If 2 or more of the 6 specimens fail to meet the specified requirements, the lot shall be rejected.

4.4.2.3.1.2 If only 1 of the 6 specimens fails to meet the the specified requirements, a second sample of 6 specimens shall be tested. Lot acceptance shall be permitted only if in the combined sample of 12 specimens not more than one specimen fails to meet the requirements. Failure of 2 or more specimens in the combined first and second samples shall result in lot rejection.

4.5 Tests.

4.5.1 *Nonmilitary.* The cable and wire shall be subject to the tests set forth in Underwriters' Laboratories, Inc. Standard UL 719 to determine compliance with 3.1.

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4.5.2 *Military.* The cable and wire shall be subject to the tests set forth in Underwriters' Laboratories, Inc. Standard UL 719, modified by the instructions set forth in 4.4, to determine compliance with 3.1.

4.6 Compliance criteria, nonmilitary.

4.6.1 *Inspection.* If one or more of the packages, 4.2.2, fails to conform to the requirements for which inspection was made, 4.3, the lot from which the packages were taken shall be considered to have failed to meet the requirements of this specification.

4.6.2 *Tests.* Compliance of the lot with respect to tests shall be based on the principles described in section 6 of J-C-98.

4.6.2.1 Any reel, coil, or roll of cable or wire which shows the presence of a defect or fails to meet the requirements when subjected to the electrical tests shall be considered to have failed to meet the requirements of this specification. If after the defects are repaired and the reel, coil, or roll is again subjected to the dielectric strength test, the presence of a defect is not shown, the reel, coil, or roll shall be considered to have met the requirements of this specification for dielectric strength.

4.7 Compliance criteria, military.

4.7.1 The compliance criteria shall be as set forth in 4.4.

5. PREPARATION FOR DELIVERY

For civil agency procurement, Fed. Std. No. 102 shall be referred to for definitions and applications of the various levels of packaging and packing protection for supplies and equipment.

5.1 Packaging.

5.1.1 *Levels A and B.*

5.1.1.1 *Coils (AWG sizes 14 to AWG 10).* Cable and wire shall be uniformly coiled and packaged 250 feet in a fiberboard box conforming to PPP-B-636, type I or II, class 2, provided with special knock-out feature normally used in commercial practice for dispensing purposes.

5.1.1.2 *Coils (AWG sizes 8 to AWG 2).* Cable and wire shall be uniformly coiled and packaged 250 feet per coil. The coils shall be spirally wrapped with waterproofed kraft paper conforming to UU-P-271, class E-2 and sealed.

5.1.1.3 *Reels.* Cable and wire shall be packaged on wood or metal reels or spools suitably designed with strong flanges of sufficient diameter to protect the cable or wire from damage during shipment and storage. After the cable or wire has been wound on the reel, it shall be covered with a full wrap of waterproofed kraft barrier material conforming to UU-P-271, class E-2. The wrap shall extend between the flanges and shall be secured in place with a minimum of two flat steel straps.

5.1.2 *Level C.* Cable and wire shall be packaged in coils or on reels of the size and kind commonly used which will afford the degree of protection required for shipment and use of the product for its intended purpose within the continental limits of the United States of America.

5.1.3 *Military agencies—Level A.* The cable and wire shall be packaged in accordance with level A of MIL-C-12000.

5.2 Packing.

5.2.1 *Level A.*

5.2.1.1 *Coils (AWG sizes 14 to AWG 10).* Cable and wire packaged as specified in 5.1.1.1 shall be packed in nailed wood or wood-cleated plywood boxes conforming to PPP-B-621 and PPP-B-601, respectively. Containers shall be fitted with waterproof case liners conforming to MIL-L-10547, class 2 and sealed in accordance with appendix thereto.

5.2.1.2 *Coils (AWG sizes 8 to AWG 2) and reels.* Cable and wire packaged as speci-

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fied in 5.1.1.2 and 5.1.1.3 shall require no further overpacking.

5.2.2 *Level B.*

5.2.2.1 *Coils and reels.* Cable and wire packaged as specified in 5.1.1.1, 5.1.1.2 or 5.1.1.3 shall require no further overpacking.

5.2.3 *Level C.* The subject commodity shall be packaged and packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.2.4 *Military agencies—Levels A and B.* Cable and wire shall be packed in accordance with level A or B, as specified, of MIL-C-12000.

5.3 Marking.

5.3.1 *Civil agencies.* In addition to any special marking required with the contract or order, shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 *Military agencies.* In addition to any special marking required in the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 **Ordering data.** Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(a) Title, symbol, and date of this specification.
(b) Type, class, grade, number of insulated conductors, and conductor size (see 1.2).
(c) Conductor material.
(d) Administrative provisions for inspection records (4.1).
(e) Lot formation and sampling, if other than specified (4.2).
(f) Levels of packaging and packing required (5.1 and 5.2).
(g) Marking required (5.3).

6.2 If laboratory tests by the Government are required, purchasing officers should order cable or wire for test purposes in addition to the number of feet required for installation.

6.3 **Transportation description.** Transportation descriptions and minimum weights applicable to this commodity are:

Rail: Cable, Electric, Copper, Not Otherwise Indexed By Name.
Carload minimum weight 30,000 pounds.

Motor: Cable, Electric, Copper, Not Otherwise Indexed.
Truckload minimum weight 30,000 pounds, subject to Rule 34, National Motor Freight Classification.

Rail: Cable, Electric, Aluminum.
Carload minimum weight 30,000 pounds.

Motor: Cable, Electric, Aluminum.
Truckload minimum weight 30,000 pounds, subject to Rule 34, National Motor Freight Classification.

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7. DEPARTMENTAL REQUIREMENTS

7.1 The following specifications and standards of the issues in effect on date of invitation for bids, and special requirements, form a part of this specification for purchases made under this specification by the respective departments.

7.2 Army, Navy, and Air Force.

7.2.1 *Applicable specifications and standard:*

Military Specification:

MIL–P–12000—Packaging and Packing of Cable, Cord and Wire, Electric.

Military Standard:

MIL–STD–129—Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

7.2.2 *Packaging for domestic shipment, storage, and overseas shipment.*—The cable shall be packaged in conformance with Military Specification MIL–P–12000.

7.2.3 *Packing for domestic shipment, storage, and overseas shipment.*—The packaged cable shall be packed in conformance with Military Specification MIL–P–12000.

7.2.4 *Marking.*—In addition to any special marking required, packages and shipping containers shall be marked in accordance with Military Standard MIL–STD–129.

MILITARY INTEREST:
Army—E M Sig T
Navy—MC S Y
Air Force

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